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# Online Examination Report

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|----------------|-----------------------|
| Date:          | 20.08.2014            |
| Attempts:      | 1                     |
| Examinee:      | Kaija, Albert         |
| Date of birth: |                       |
| Subject:       | 040-Human performance |
| Result:        | 60 %                  |
| Time used:     | 00:18:15 (HH:MM:SS)   |

## Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

## Attachment 1: List of result by question

| CQB Number | Internal Number | Syllabus reference | Chapter                                          | Points | Max. Points |
|------------|-----------------|--------------------|--------------------------------------------------|--------|-------------|
| 1          | 204             | 40.                | HUMAN PERFORMANCE                                | 1,00   | 1,00        |
| 2          | 202             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 3          | 200             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 4          | 209             | 40.                | HUMAN PERFORMANCE                                | 1,00   | 1,00        |
| 5          | 162             | 40.                | HUMAN PERFORMANCE                                | 1,00   | 1,00        |
| 6          | 192             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 7          | 171             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 8          | 174             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 9          | 129             | 40.                | HUMAN PERFORMANCE                                | 0,00   | 1,00        |
| 10         | 108             | 40.2               | Basic aviation physiology and health maintenance | 0,00   | 1,00        |
| 11         | 103             | 40.2               | Basic aviation physiology and health maintenance | 0,00   | 1,00        |
| 12         | 68              | 40.2.1.1           | The atmosphere                                   | 1,00   | 1,00        |
| 13         | 55              | 40.2.1.1           | The atmosphere                                   | 1,00   | 1,00        |
| 14         | 99              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 15         | 77              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 16         | 12              | 40.2.1.2           | Respiratory and circulatory systems              | 0,00   | 1,00        |
| 17         | 60              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 18         | 47              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 19         | 38              | 40.2.1.2           | Respiratory and circulatory systems              | 1,00   | 1,00        |
| 20         | 102             | 40.2.2             | Man and Environment: the sensory system          | 0,00   | 1,00        |
| 21         | 41              | 40.2.2.2           | Vision                                           | 1,00   | 1,00        |
| 22         | 50              | 40.2.2.2           | Vision                                           | 0,00   | 1,00        |
| 23         | 42              | 40.2.2.2           | Vision                                           | 1,00   | 1,00        |
| 24         | 62              | 40.2.2.3           | Hearing                                          | 1,00   | 1,00        |
| 25         | 85              | 40.2.2.5           | Integration of sensory inputs                    | 0,00   | 1,00        |
| 26         | 107             | 40.2.3             | Health and hygiene                               | 1,00   | 1,00        |
| 27         | 4               | 40.2.3.3           | Problem areas for pilots                         | 1,00   | 1,00        |
| 28         | 43              | 40.2.3.3           | Problem areas for pilots                         | 1,00   | 1,00        |

| CQB Number       | Internal Number | Syllabus reference | Chapter                        | Points       | Max. Points  |
|------------------|-----------------|--------------------|--------------------------------|--------------|--------------|
| 29               | 31              | 40.2.3.4           | Intoxication                   | 0,00         | 1,00         |
| 30               | 14              | 40.3               | BASIC AVIATION PSYCHOLOGY      | 1,00         | 1,00         |
| 31               | 121             | 40.3               | BASIC AVIATION PSYCHOLOGY      | 0,00         | 1,00         |
| 32               | 144             | 40.3.1             | Human information processing   | 1,00         | 1,00         |
| 33               | 97              | 40.3.1.1           | Attention and vigilance        | 1,00         | 1,00         |
| 34               | 80              | 40.3.1.3           | Memory                         | 1,00         | 1,00         |
| 35               | 40              | 40.3.2.1           | Reliability of human behaviour | 0,00         | 1,00         |
| 36               | 21              | 40.3.2.4           | Error generation               | 1,00         | 1,00         |
| 37               | 81              | 40.3.3.1           | Decision-making concepts       | 1,00         | 1,00         |
| 38               | 5               | 40.3.4.3           | Co-operation                   | 1,00         | 1,00         |
| 39               | 123             | 40.3.5             | Human behaviour                | 1,00         | 1,00         |
| 40               | 7               | 40.3.6.1           | Arousal                        | 1,00         | 1,00         |
| <b>Total:60%</b> |                 |                    |                                | <b>24,00</b> | <b>40,00</b> |

## Attachment 2: List of result by topic

**Licence**

**Date: 20.08.2014**

**Examinee: Kaija, Albert**

**Location: Uganda Civil Aviation Authority**

**Your Result: 24,00 from 40,00 Points (60,00 %)**

| Topics                                                 | Ammount of Questions | Points achieved | Maximum Points | Percent      |
|--------------------------------------------------------|----------------------|-----------------|----------------|--------------|
| 40. HUMAN PERFORMANCE                                  | 40                   | 24,00           | 40,00          | 60,00        |
| 40.2. Basic aviation physiology and health maintenance | 20                   | 12,00           | 20,00          | 60,00        |
| 40.3. BASIC AVIATION PSYCHOLOGY                        | 11                   | 9,00            | 11,00          | 81,82        |
| <b>Total</b>                                           | <b>40</b>            | <b>24,00</b>    | <b>40,00</b>   | <b>60,00</b> |

1

**A pilot is more subject to spatial disorientation if**

- ☐ **kinesthetic senses are ignored.**
- ☒ **body signals are used to interpret flight attitude.**
- ☐ **eyes are moved often in the process of cross-checking the flight instruments.**

2

Which statement is correct regarding the use of cockpit lighting for night flight?

- ☐ The use of regular white light, such as a flashlight, will impair night adaptation.
- ☒ Reducing the lighting intensity to a minimum level will eliminate blind spots.
- ☐ Coloration shown on maps is least affected by the use of direct red lighting.

3

**The sensations which lead to spatial disorientation during instrument flight conditions**

- ☐ must be suppressed and complete reliance placed on the indications of the flight instruments.
- ☒ occur, in most instances, during the initial period of transition from visual to instrument flight.
- ☐ are frequently encountered by beginning instrument pilots, but never by pilots with moderate instrument experience.

4

**What effect does haze have on the ability to see traffic or terrain features during flight?**

- ☐ Haze causes the eyes to focus at infinity, making terrain features harder to see.
- ☒ Haze creates the illusion of being a greater distance than actual from the runway, and causes pilots to fly a lower approach.
- ☐ The eyes tend to overwork in haze and do not detect relative movement easily.



5

**What physical change would most likely occur to occupants of an unpressurized aircraft flying above 15,000 feet without supplemental oxygen?**

- ☐ The pressure in the middle ear becomes less than the atmospheric pressure in the cabin.
- ☒ A blue coloration of the lips and fingernails develop along with tunnel vision.
- ☐ Gases trapped in the body contract and prevent nitrogen from escaping the bloodstream.

6

Without visual aid, a pilot often interprets centrifugal force as a sensation of

☒ turning.

☐ rising or falling.

☐ motion reversal.

7

**During training flights, an instructor should interject realistic distractions to determine if a student can**

- ☒ maintain aircraft control while his/her attention is diverted.
- ☐ learn despite stressful conditions.
- ☒ perform maneuvers using the integrated method of flight instruction.

8

**One of the steps in the aeronautical decision making (ADM) process involved in good decision making is**

- ☒ making a rational evaluation of the required actions.
- ☐ developing a 'can do' attitude.
- ☐ identifying personal attitudes hazardous to safe flight.

9

**Which of the following is the final step of the Decide Model for effective risk management and Aeronautical Decision Making?**

☐ **Estimate.**

☒ **Eliminate.**

☐ **Evaluate.**

10

**Hypoxia susceptibility due to inhalation of carbon monoxide increases as**

- ☐ altitude increases.
- ☒ oxygen demand increases.
- ☐ humidity decreases.

11

Which of these is a common symptom of hyperventilation?

- ☒ Drowsiness.
- ☐ A sense of well-being.
- ☒ Decreased breathing rate.

12

**Fatigue and stress**

- ☐ increase the tolerance to hypoxia
- ☐ will increase the tolerance to hypoxia when flying below 15 000 feet
- ☒ lower the tolerance to hypoxia
- ☐ do not affect hypoxia at all



13

**Air at an altitude of 18.000 feet contains, approximately:**

☐ 10% oxygen

☒ 21% oxygen

☐ 15% oxygen

☐ 5% oxygen

14

**Hypoxia can be caused by**

- ☐ lack of nitrogen in ambient air
- ☐ increasing oxygen partial pressure used for the exchange of gases
- ☒ decreased ability of the haemoglobin to transport oxygen
- ☐ too much carbon dioxide in the blood

15

**During a rapid decompression in a 2 crew aircraft, the other pilot becomes incapacitated, your first action is to:**

- ☐ call for immediate assistance from cabin crew.
- ☐ administer oxygen to the other pilot.
- ☒ commence an emergency descent.
- ☐ don your own oxygen mask.

16

**Hypoxia is caused by**

- ☐ an increased number of red blood cells
- ☐ reduced partial pressure of nitrogen in the lung
- ☒ reduced partial oxygen pressure in the lung
- ☒ a higher affinity of the red blood cells (haemoglobin) to oxygen

17

The following actions are appropriate when faced with symptoms of decompression sickness:

1. climb to higher level
2. descent to the higher of 10000 ft or MSA and land as soon as possible
3. breathe 100 % oxygen
4. obtain medical advice about recompression after landing

☒ 2, 3 and 4 are correct

☐ 1 and 4 are correct

☐ 1 and 3 are correct

☐ 1, 2 and 3 are correct

18

**The respiratory process consists mainly of**

- ☐ the transportation of oxygen to the cells and the elimination of carbon monoxide
- ☐ the transportation of carbon dioxide to the cells and elimination of oxygen
- ☐ the transportation of oxygen to the cells and the elimination of nitrogen
- ☒ the diffusion of oxygen through the respiratory membranes into the blood, transportation to the cells, diffusion into the cells and elimination of carbon dioxide from the body

19

**Flying a non-pressurized light aircraft at 9 000 feet your passenger develops blue lips:**

- ☐ you supply him/her with extra oxygen, but do not reduce altitude .
- ☐ calm him/her and occupy him with suitable tasks.
- ☒ you reduce altitude and supply him/her with oxygen.
- ☐ you supply additional oxygen but controlled.

20

**To scan properly for traffic, a pilot should**

- ☐ **use a series of short, regularly spaced eye movements that bring successive areas of the sky into the central visual field.**
- ☐ **concentrate on any peripheral movement detected.**
- ☒ **slowly sweep the field of vision from one side to the other at intervals.**



21

Which of the following statements is correct?

- ☐ Photopic Vision is the same as Peripheral Vision.
- ☒ Scotopic Vision is vision through the operation of the Rods.
- ☐ Scotopic Vision is the same as Photopic Vision.
- ☐ Scotopic Vision is vision through the operation of the Cones.

22

**Cataract is caused by:**

- ☐ A clouding of the lens
- ☒ A lack of accommodation at the cornea
- ☐ A mis-shapened cornea
- ☐ Lack of mobility of the cornea

23

When flying through a thunderstorm with lightning you can protect yourself from flashblindness by:

- 1 - turning up the intensity of cockpit lights
- 2 - looking inside the cockpit
- 3 - wearing sunglasses
- 4 - using blinds or curtains when installed

☐ 1, 2 and 3 are correct, 4 is false

☐ 3 and 4 are correct, 1 and 2 are false

☒ 1, 2, 3 and 4 are correct

☐ 1 and 2 are correct, 3 and 4 are false

24

Which part of the inner ear is responsible for the perception of sound?

- ☒ The cochlea
- ☐ The sacculus and utriculus
- ☐ The Eustachian tube
- ☐ The semicircular canals

25

**"Pilot's vertigo"**

- ☒ is the sensation of climbing caused by a strong linear acceleration
- ☐ is the sensation to keep a rotation after completing a turn
- ☐ is the condition of dizziness and/or tumbling sensation caused by contradictory impulses to the central nervous system (CNS)
- ☐ announces the beginning of airsickness

26

Which of these is true regarding the presence of alcohol within the human body?

- ☐ An increase in altitude decreases the adverse effect of alcohol.
- ☐ A small amount of alcohol increases vision acuity.
- ☒ Judgment and decision-making abilities can be adversely affected by even small amounts of alcohol.

27

**How is Yellow fever contracted?**

- ☐ Contaminated food or water.
- ☐ Excessive use of alcohol.
- ☒ A virus transmitted by an infected mosquito.
- ☐ By contact with the saliva of infected animals.

28

Medical conditions such as high blood pressure, coronary problems and diabetes are associated with:

☐ hypoxia

☒ obesity

☐ cholera

☐ anorexia nervosa



29

**Alcohol, when taken simultaneously with drugs, may**

- ☒ **increase the rate of drug elimination from the blood**
- ☐ **intensify the side effects of the drugs**
- ☐ **lead to undesired effects only during night flights**
- ☐ **compensate for side effects of drugs**

30

**Carbon Monoxide is particularly dangerous because:**

1. Its initial symptoms are not alarming
2. It is colourless
3. Its is odourless
4. It is highly toxic
5. Its effects are cumulative

☒ all of the above

☐ 2, 3, and 4 only

☐ 2, 3, 4 and 5 only

☐ 1, 2, 3 and 5 only

31

**To help manage cockpit stress, pilots must**

- ☐ be aware of life stress situations that are similar to those in flying.
- ☒ avoid situations that will improve their abilities to handle cockpit responsibilities.
- ☐ condition themselves to relax and think rationally when stress appears.

32

**Before a student can concentrate on learning, which of these human needs must be satisfied first?**

☐ **Safety needs.**

☒ **Physical needs.**

☐ **Social needs.**

33

Which of the following statements are correct?

1. The first information received determines how subsequent information will be evaluated.
2. If one has made up one's mind, contradictory information may not get the attention it really needs.
3. With increasing stress, attention is limited thereby reducing the flow of information to the central decision maker.

- ☐ 1 and 3 are correct
- ☐ 2 and 3 are correct
- ☒ 1, 2 and 3 are correct
- ☐ 1 and 2 are correct

34

Working memory enables us, for example,

- ☐ to remember our own name
- ☒ to remember a clearance long enough to write it down
- ☐ to ignore messages for other aircraft
- ☐ to store a large amount of visual information for about 0.5 seconds

35

Situations particularly vulnerable to "reversion to an earlier behaviour pattern" are:

1. when concentration on a particular task is relaxed
2. when situations are characterised by medium workload
3. when situations are characterised by stress

☐ 1. and 2.

☒ 2. and 3.

☐ 1. and 3.

☐ 3 only.

36

Human error rates during the performance of a simple and repetitive task can normally be expected to be approximately:

☐ 1 in 500

☐ 1 in 200

☒ 1 in 100

☐ 1 in 2000



37

**Judgement is based upon:**

- ☐ a decision-making process involving the 5 physical senses and their use to manually operate the aircraft controls
- ☐ the ability to interpret the flight instruments
- ☐ the development of skills through constant practice of flight manoeuvres
- ☒ a process involving a pilot's attitude to take and to evaluate risks by assessing the situation and making decisions based upon knowledge, skill and experience

38

Which of the following statements best characterises a self-centered cockpit ?

- ☒ Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on
- ☐ The communication between crew members always increases when the captain takes charge of a situation
- ☐ While decreasing communication, the independence of each member bolsters the crew's synergy
- ☐ The egoistic and self-centered personality of the captain often leads to a synergetic cockpit

39

The passengers for a charter flight have arrived almost an hour late for a flight that requires a reservation. Which of the following alternatives best illustrates the ANTIAUTHORITY reaction?

- ☐ The pilot can't help it that the passengers are late.
- ☐ If the pilot hurries, he or she may still make it on time.
- ☒ Those reservation rules do not apply to this flight.

40

**An identical situation can be experienced by one pilot as exciting in a positive sense and by another pilot as threatening. In both cases:**

- ☐ both pilots will experience the same amount of stress
- ☐ both pilots will loose their motor-coordination
- ☒ the arousal level of both pilots will be raised
- ☐ the pilot feeling threatened, will be much more relaxed, than the pilot looking forward to what may happen